Software Requirements Specifications

Version 1.0

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Study Buddy Platform

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Revision History

Revision Letter	Primary Author	Description	Date
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1. Introduction

1.1 Scope

In-scope: Initially, the application would be available only to CSUN students for their classes taken at the university. The release will prioritize a reliable and user-friendly platform for students to share their interests and career goals. The intended audience would be CSUN students seeking a study group.

Out of scope: Beyond the initial release, the application will be extended to other college campuses. Native apps for IOS or Android would also be developed. Seamless integration into Canvas or the CSUN portal would also be a long-term goal. Eventually, the web application would be released to the general public.

1.2 Product Value

At CSUN, students face challenges coordinating study sessions, finding reliable peer support, and academic engagement.

The Study Buddy Platform addresses these challenges by centralizing functions such as, study room reservations, peer to peer matching, and accessibility to class specific forums into a user friendly hub.

1.3 Intended Audience

Study Buddy is a platform designed to be used for all types of learners, but with a main emphasis on university students.

1.4 Intended Use

For the purposes of this course, Study Buddy is an application by CSUN students, for CSUN students. However, it aims to implement a scalable design to be potentially used by other universities and even non-academic settings.

Functional Requirements

1.5 General Requirements

Requirement	Description
GR_SRS_001	The platform shall allow users to register and authenticate profiles.
GR_SRS_002	The platform shall securely store user data using hashing methods.
GR_SRS_003	The platform shall allow users to chat in real time.
GR_SRS_004	The platform shall recommend and match users to each other based on profile similarity.
GR_SRS_005	The platform shall allow users to form and join study groups.
GR_SRS_006	The platform shall feature a forum for users to engage in discussion.
GR_SRS_007	The platform shall integrate Google OAuth 2.0.
GR_SRS_008	The platform shall be supported on all major platforms (Chrome, Firefox, Safari, etc).
GR_SRS_009	The platform shall store profile, study group, chat, and other data in a MongoDB database.

1.6 User Requirements

Requirement	Description
USER_SRS_001	The user shall be able to find compatible study buddies based on similar subjects and courses.
USER_SRS_002	The user shall be able to send out connection requests to other people.
USER_SRS_003	The user shall be able to view suggested people to connect with.

USER_SRS_004	The user shall be able to send direct or group messages to other users.
USER_SRS_005	The user shall be able to create or join study sessions.
USER_SRS_006	The user shall be notified of upcoming study sessions.
USER_SRS_007	The user shall be able to block and report inappropriate users.

1.7 Frontend Requirements

Requirement	Description
FRT_SRS_001	The frontend shall provide a login field for CSUN email and a password field.
FRT_SRS_002	The frontend shall validate the login credentials from the backend authentication.
FRT_SRS_003	The frontend shall display an error message if authentication fails.
FRT_SRS_004	The frontend shall direct the user to the dashboard upon successful authentication.
FRT_SRS_005	The frontend shall block user registration if the required fields are not approved.
FRT_SRS_006	The frontend shall display a feed page (user and study group posts, profile recommendations, and messaging feature)
FRT_SRS_007	The frontend shall display notifications from the messages or feed page for the user.
FRT_SRS_008	The frontend shall display the user's profile with a customizing page feature.
FRT_SRS_009	The frontend shall provide a browsable group explore page that allows filtering by interests.
FRT_SRS_010	The frontend shall provide a page displaying other users with connect buttons that allow friend requests to be sent to each other.
FRT_SRS_011	The frontend shall display incoming and outgoing connection

	requests that allow users to accept or deny them.
FRT_SRS_012	The frontend shall provide a chatbox interface that shows ongoing conversations.
FRT_SRS_013	The frontend shall provide a forum page where users can post, view, and respond to each other.
FRT_SRS_014	The frontend shall store and access environment variables in a .env file.
FRT_SRS_015	The interface shall function properly across different screen sizes.

1.8 Backend Requirements

Requirement	Description
BCK_SRS_001	The backend shall provide a configured development environment using FastAPI and a package manager.
BCK_SRS_002	The backend shall support project scripts for development, build, and production execution.
BCK_SRS_003	The backend shall load configuration securely from environment variables (e.g., .env).

2. External Interface Requirements

2.1 User Interface Requirements

Requirement	Description
UI_SRS_001	The user shall be able to securely login using their CSUN email and password.
UI_SRS_002	The user shall be able to edit their profile information.
UI_SRS_003	The user shall be able to specify and display their courses and study habits.
UI_SRS_004	The user shall be able to choose what parts of their profile are visible.

2.2 Software Interface Requirements

Requirement	Description
SI_SRS_001	The system shall establish a connection to MongoDB using a secure connection string.
SI_SRS_002	The backend shall expose a health-check or logging mechanism to check MongoDB connectivity.
SI_SRS_003	The system shall handle database connection failures and log descriptive error messages.
SI_SRS_004	The backend shall implement a User schema in MongoDB.
SI_SRS_005	The user schema shall include required fields such as email, name, and timestamps.
SI_SRS_006	The system shall enforce unique email addresses for all users.
SI_SRS_007	The system shall validate user data before insertion or updates.
SI_SRS_008	The backend shall implement Inngest webhook endpoints for user creation.
SI_SRS_009	The backend shall implement Inngest webhook endpoints for

	user updates.
SI_SRS_010	The backend shall implement Inngest webhook endpoints for user deletion.
SI_SRS_011	All webhooks shall verify request authenticity (e.g., signature/secret).
SI_SRS_012	Webhook requests shall be idempotent and safe against duplicate events.
SI_SRS_013	The system shall return appropriate HTTP status codes for webhook outcomes.

2.3 Communication Interface Requirements

Requirement	Description
CI_SRS_001	The web browser and the server will communicate over HTTPS protocol
CI_SRS_002	The server will authenticate user authorization to modify data using JWT tokens
CI_SRS_003	The backend will include CORS headers to ensure that all requests come from the Study Buddy's website domain

3. Non-Functional Requirements

3.1 Security

Requirement	Description
SEC_SRS_001	The system shall only allow authorized users to access sensitive data.
SEC_SRS_002	The system shall hash and salt passcodes when storing them.

3.2 Capacity

Requirement	Description
CAP_SRS_001	The system shall support more than one concurrent user at a time.
CAP_SRS_002	Must be able to support a number of user profiles, with ability to expand additional users.
CAP_SRS_003	The database must store basic user profile info, course information provided, and optimized for future expansions.
CAP_SRS_004	Must scale to additional campuses for later phases without major need for redesign in architecture.

3.3 Compatibility

Requirement	Description
COM_SRS_001	The website shall be compatible with Google Chrome, Safari, Firefox, and Edge.

3.4 Reliability

Requirement	Description

REL_SRS_001	The system shall give an error message when a failure happens and then recover automatically.
REL_SRS_002	The system shall have 99% uptime, with downtime only for scheduled maintenance.
REL_SRS_003	The system shall not corrupt user data in the event of a crash

3.5 Scalability

Requirement	Description
SCL_SRS_001	The system shall be scalable to any number of users, limited only by the database capacity and funds.

3.6 Usability

Requirement	Description
USE_SRS_001	The website shall be usable on both desktop and mobile devices.

Qualification Provisions

SRS Req. ID	Description	Verification Method
GR_SRS_001	The platform shall allow users to register and authenticate profiles.	Demonstration
GR_SRS_002	The platform shall securely store user data using hashing methods.	Analysis
GR_SRS_003	The platform shall allow users to chat in real time.	Demonstration
GR_SRS_004	The platform shall recommend and match users to each other based on profile similarity.	Analysis
GR_SRS_005	The platform shall allow users to form and join study groups.	Demonstration
GR_SRS_006	The platform shall feature a forum for users to engage in discussion.	Demonstration
GR_SRS_007	The platform shall integrate Google OAuth 2.0.	Inspection
GR_SRS_008	The platform shall be supported on all major platforms (Chrome, Firefox, Safari, etc).	Demonstration
GR_SRS_009	The platform shall store profile, study group, chat, and other data in a MongoDB database.	Inspection
USER_SRS_001	The user shall be able to find compatible study buddies based on similar subjects and courses.	Demonstration

USER_SRS_002	The user shall be able to send out connection requests to other people.	Demonstration
USER_SRS_003	The user shall be able to view suggested people to connect with.	Demonstration
USER_SRS_004	The user shall be able to send direct or group messages to other users.	Demonstration
USER_SRS_005	The user shall be able to create or join study sessions.	Demonstration
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USER_SRS_007	The user shall be able to block and report inappropriate users.	Demonstration
FRT_SRS_001	The frontend shall provide a login field for CSUN email and a password field.	Inspection
FRT_SRS_002	The frontend shall validate the login credentials from the backend authentication.	Test
FRT_SRS_003	The frontend shall display an error message if authentication fails.	Test
FRT_SRS_004	The frontend shall direct the user to the dashboard upon successful authentication.	Test
FRT_SRS_005	The frontend shall block user registration if the required fields are not approved.	Test
FRT_SRS_006	The frontend shall display a feed page (user and study group posts, profile recommendations, and messaging feature)	Demonstration

FRT_SRS_007	The frontend shall display notifications from the messages or feed page for the user.	Demonstration
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FRT_SRS_015	The interface shall function properly across different screen sizes.	Demonstration
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USE_SRS_001	The website shall be usable on both desktop and mobile devices.	Demonstration